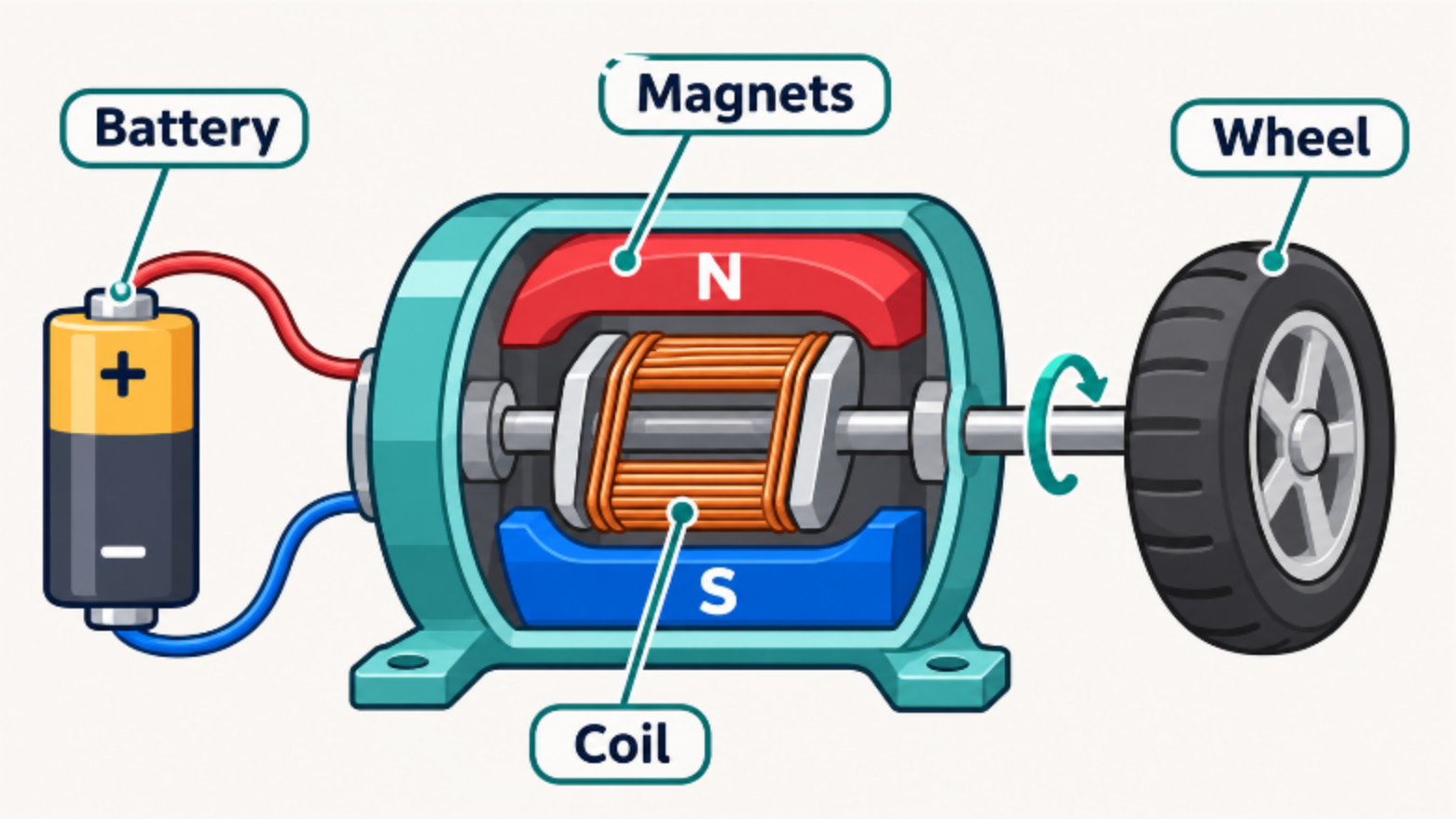
Delivery to a target

How wheel turns carry our delivery 40 cm

What makes the robot move



Electricity flows through the coil.

Magnetic forces turn the motor shaft and the wheel.

The tire grips the floor and moves the robot forward.

Simplified motor picture. The hub controls the motors on our robot.

Degrees measure a wheel turn



90°

Quarter turn



180°

Half turn

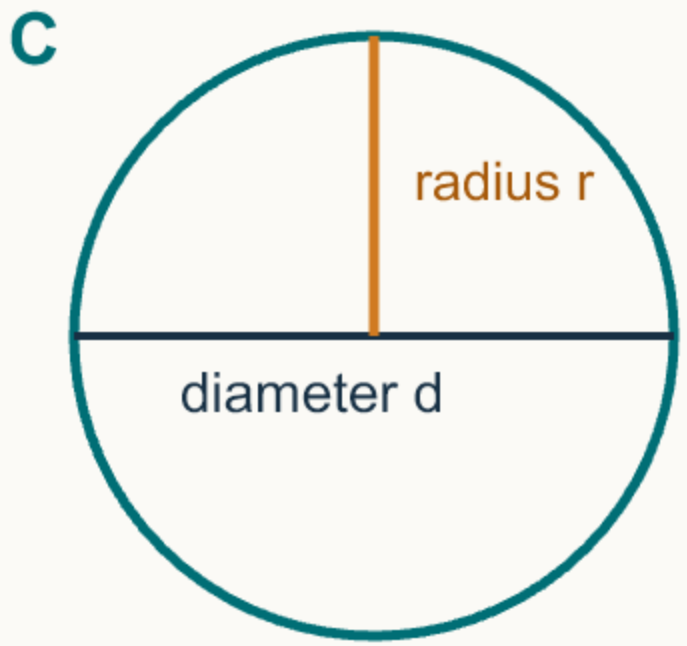


360°

Full turn

Wheel degrees describe how much the wheels rotate.

Circumference is the distance around



$$C = \pi \times d$$

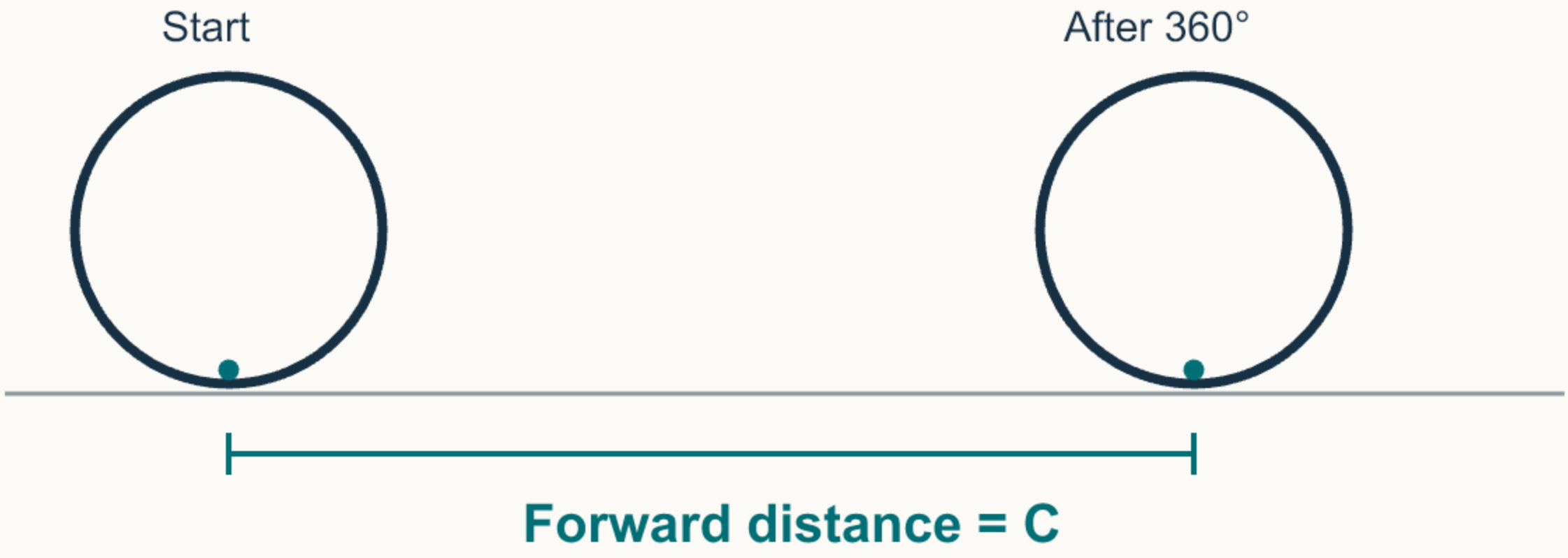
$$C = 2 \times \pi \times r$$

π is about 3.14

Example: d = 5.6 cm
C ≈ 3.14 × 5.6 ≈ 17.6 cm

Measure the actual tire on each kit. Your wheel size may differ.

One turn rolls one circumference



The dot returns to the same place after one full turn.

Prediction for straight rolling without slipping.

Wheel degrees to forward distance

Distance = (wheel degrees ÷ 360) × C

Example circumference C = 17.6 cm

Wheel degrees	Turns	Predicted distance
180°	½	8.8 cm
360°	1	17.6 cm
720°	2	35.2 cm

For OUR kit, how many centimeters should 180° travel?

Target distance to wheel degrees

Wheel degrees = (target distance ÷ C) × 360

Example: target 40 cm, circumference 17.6 cm

40 ÷ 17.6 ≈ 2.27 turns

(40 ÷ 17.6) × 360 ≈ 818°

Use our kit’s measured C. Round only the final answer.

A meter is 100 cm. Keep target distance and C in the same unit.

Tell the code how far the wheels turn

```
wheel_degrees = 180
await drive(wheel_degrees)
await signal("D")
```

Our prediction

Using our C, how many centimeters for 180°? For 360°?

What wheel_degrees value should aim for 40 cm?

Recipe fragment: load the complete supplied Week 2 program

Our robot and build jobs

Driver edits and starts after “clear.”

Navigator predicts and resets the stopped robot.

Observer watches, measures and records.

Other team builds. Everyone rotates jobs.

Robot turns: rotate every 4 minutes. Retests: every 2 minutes.

Compare prediction with measurement

We set wheel_degrees to _____.

We predicted _____ cm and measured _____ cm.

Our 40 cm run was _____ cm too short / too far.

What one change should we test next?

Explain the motion

What makes the motor turn?

How far should half a wheel turn go?

How would you choose the code number for a target?

Save your team project. Store your build. Reset the robot and parts.